

## Low ESR Cap.Compatible Positive Voltage Regulators

## ■ GENERAL DESCRIPTION

The XC6206 series are highly precise, low power consumption, 3 terminal, positive voltage regulators manufactured using CMOS and laser trimming technologies. The series provides large currents with a significantly small dropout voltage.

The XC6206 consists of a current limiter circuit, a driver transistor, a precision reference voltage and an error correction circuit. The series is compatible with low ESR ceramic capacitors. The current limiter's foldback circuit operates as a short circuit protection as well as the output current limiter for the output pin.

Output voltages are internally by laser trimming technologies. It is selectable in 0.1V increments within a range of 1.2V to 5.0V.

SOT-23, SOT-89, TO-92 and USP-6B packages are available.

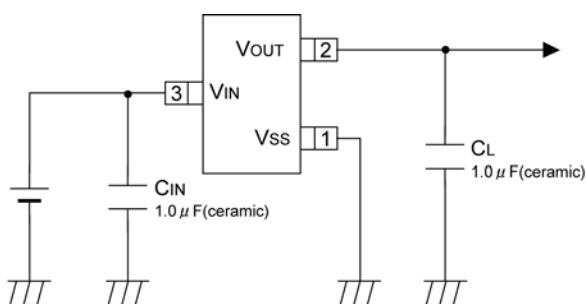
## ■ APPLICATIONS

- Smart phones / Mobile phones
- Portable game consoles
- Digital still cameras / Camcorders
- Digital audio equipments
- Reference voltage sources
- Multi-function power supplies

## ■ FEATURES

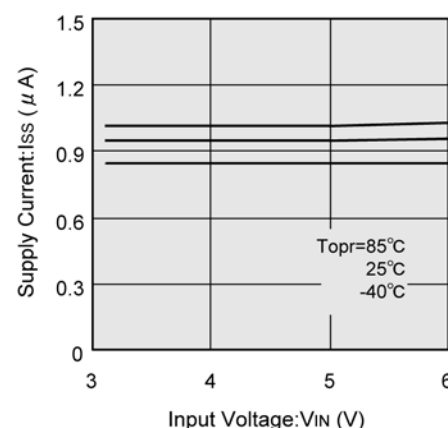
|                                      |                                                                                                                |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Maximum Output Current</b>        | : 200mA (3.0V type)                                                                                            |
| <b>Dropout Voltage</b>               | : 250mV @ 100mA (3.0V type)                                                                                    |
| <b>Maximum Operating Voltage</b>     | : 6.0V                                                                                                         |
| <b>Output Voltage Range</b>          | : 1.2V ~ 5.0V (0.1V increments)                                                                                |
| <b>Highly Accurate</b>               | : $\pm 2\%$ @ $V_{OUT} \geq 1.5V$<br>$\pm 30mV$ @ $V_{OUT} < 1.5V$<br>$(\pm 1\% \text{ @ } V_{OUT} \geq 2.0V)$ |
| <b>Low Power Consumption</b>         | : $1.0\mu A$ (TYP.)                                                                                            |
| <b>Low ESR Capacitor</b>             | : Ceramic capacitor compatible                                                                                 |
| <b>Protection</b>                    | : Current Limit Circuit Built-in                                                                               |
| <b>Operating Ambient Temperature</b> | : $-40^{\circ}C \sim +85^{\circ}C$                                                                             |
| <b>Packages</b>                      | : SOT-23<br>SOT-89<br>TO-92<br>USP-6B                                                                          |
| <b>Environmentally Friendly</b>      | : EU RoHS Compliant, Pb Free                                                                                   |

## ■ TYPICAL APPLICATION CIRCUIT

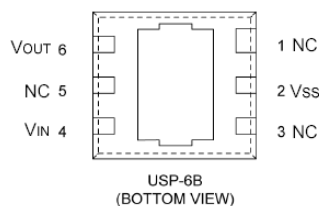
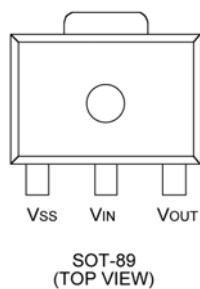
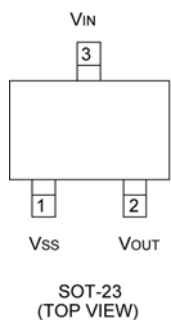


## ■ TYPICAL PERFORMANCE CHARACTERISTICS

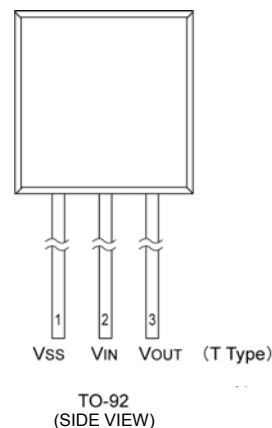
XC6206P302



## PIN CONFIGURATION



\*The dissipation pad for the USP-6B package should be solder-plated in recommended mount pattern and metal masking so as to enhance mounting strength and heat release. If the pad needs to be connected to other pins, it should be connected to the pin number 4 ( $V_{IN}$ ).



## PIN ASSIGNMENT

| PIN NUMBER |        |         |       | PIN NAME | FUNCTIONS     |
|------------|--------|---------|-------|----------|---------------|
| SOT-23     | SOT-89 | USP-6B  | TO-92 |          |               |
| 1          | 1      | 2       | 1     | VSS      | Ground        |
| 3          | 2      | 4       | 2     | VIN      | Power Input   |
| 2          | 3      | 6       | 3     | VOUT     | Output        |
| -          | -      | 1, 3, 5 | -     | NC       | No Connection |

## PRODUCT CLASSIFICATION

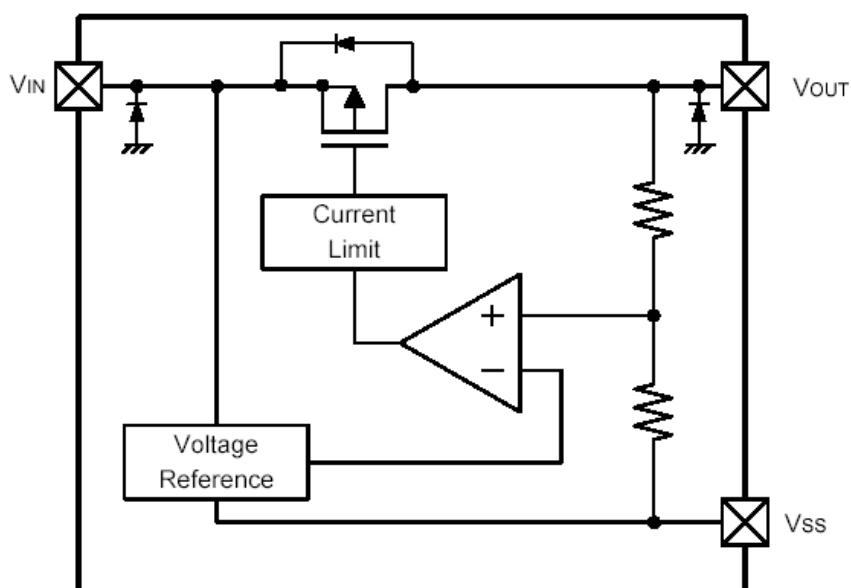
### ●Ordering Information

XC6206P ①②③④⑤-⑥<sup>(\*)</sup>

| DESIGNATOR | ITEM                     | SYMBOL | DESCRIPTION                                                     |
|------------|--------------------------|--------|-----------------------------------------------------------------|
| ①②         | Output Voltage           | 12~50  | e.g. V <sub>OUT</sub> : 3.0V→①=3, ②=0                           |
| ③          | Accuracy                 | 2      | ± 2% (V <sub>OUT</sub> ≥ 1.5V), ±30mV (V <sub>OUT</sub> < 1.5V) |
|            |                          | 1      | ±1% (V <sub>OUT</sub> ≥ 2.0V)                                   |
| ④⑤-⑥       | Packages<br>(Order Unit) | MR     | SOT-23 (3,000/Reel)                                             |
|            |                          | MR-G   | SOT-23 (3,000/Reel)                                             |
|            |                          | PR     | SOT-89 (1,000/Reel)                                             |
|            |                          | PR-G   | SOT-89 (1,000/Reel)                                             |
|            |                          | DR     | USP-6B (3,000/Reel)                                             |
|            |                          | DR-G   | USP-6B (3,000/Reel)                                             |
|            |                          | TH     | TO-92 (T type), Paper type (2,000/Tape)                         |
|            |                          | TH-G   | TO-92 (T type), Paper type (2,000/Tape)                         |
|            |                          | TB     | TO-92 (T type), Bag type (500/Bag)                              |
|            |                          | TB-G   | TO-92 (T type), Bag type (500/Bag)                              |

<sup>(\*)</sup> The "-G" suffix denotes Halogen and Antimony free as well as being fully EU RoHS compliant.

## ■ BLOCK DIAGRAM



\*Diodes inside the circuit are an ESD protection diode and a parasitic diode.

## ■ ABSOLUTE MAXIMUM RATINGS

Ta=25°C

| PARAMETER                     |        | SYMBOL           | RATINGS                                       | UNITS |
|-------------------------------|--------|------------------|-----------------------------------------------|-------|
| Input Voltage                 |        | V <sub>IN</sub>  | 7.0                                           | V     |
| Output Current                |        | I <sub>OUT</sub> | 500 *                                         | mA    |
| Output Voltage                |        | V <sub>OUT</sub> | V <sub>SS</sub> - 0.3 ~ V <sub>IN</sub> + 0.3 | V     |
| Power Dissipation             | SOT-23 | P <sub>d</sub>   | 250                                           | mW    |
|                               | SOT-89 |                  | 500                                           |       |
|                               | USP-6B |                  | 100                                           |       |
|                               | TO-92  |                  | 300                                           |       |
| Operating Ambient Temperature |        | T <sub>opr</sub> | - 40 ~ + 85                                   | °C    |
| Storage Temperature           |        | T <sub>stg</sub> | - 55 ~ + 125                                  | °C    |

\* I<sub>OUT</sub>=P<sub>d</sub> / (V<sub>IN</sub>-V<sub>OUT</sub>)

## ELECTRICAL CHARACTERISTICS

### XC6206P series

Ta=25 °C

| PARAMETER                                  | SYMBOL                                                | CONDITIONS <sup>(*)</sup>                                                                                                                                           | MIN.   | TYP.                                | MAX.   | UNIT<br>S  | CIRCUIT |
|--------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------|--------|------------|---------|
| Output Voltage <sup>(*)4</sup>             | V <sub>OUT(E)</sub> <sup>(*)3</sup>                   | I <sub>OUT</sub> =30mA                                                                                                                                              | x 0.98 | V <sub>OUT(T)</sub> <sup>(*)2</sup> | x 1.02 | V          | ①       |
|                                            |                                                       |                                                                                                                                                                     | E-1    |                                     |        |            |         |
| Maximum Output Current                     | I <sub>OUTMAX</sub>                                   | -                                                                                                                                                                   | E-2    | -                                   | -      | mA         | ①       |
| Load Regulation                            | ΔV <sub>OUT</sub>                                     | V <sub>OUT(T)</sub> >1.8V: 1mA ≤ I <sub>OUT</sub> ≤ 100mA<br>V <sub>OUT(T)</sub> ≤1.8V: 1mA ≤ I <sub>OUT</sub> ≤ 50mA                                               | -      | -                                   | E-3    | mV         | ①       |
| Dropout Voltage <sup>(*)5</sup>            | V <sub>dif1</sub>                                     | I <sub>OUT</sub> =30mA                                                                                                                                              | -      | E-4                                 |        | mV         | ①       |
|                                            | V <sub>dif2</sub>                                     | V <sub>OUT(T)</sub> >1.8V: I <sub>OUT</sub> =100mA<br>V <sub>OUT(T)</sub> ≤1.8V: I <sub>OUT</sub> =60mA                                                             | -      | E-5                                 |        | mV         |         |
| Supply Current                             | I <sub>DD</sub>                                       | V <sub>CE</sub> =V <sub>IN</sub>                                                                                                                                    | -      | 1.0                                 | 3.0    | μA         | ②       |
| Line Regulation                            | $\frac{\Delta V_{OUT}}{\Delta V_{IN} \cdot V_{OUT}}$  | V <sub>OUT(T)</sub> <4.5V: V <sub>OUT(T)</sub> +1.0V ≤ V <sub>IN</sub> ≤ 6.0V<br>V <sub>OUT(T)</sub> ≥4.5V: 5.5V ≤ V <sub>IN</sub> ≤ 6.0V<br>I <sub>OUT</sub> =30mA | -      | 0.05                                | 0.25   | %/V        | ①       |
| Input Voltage                              | V <sub>IN</sub>                                       | -                                                                                                                                                                   | 1.8    | -                                   | 6.0    | V          | -       |
| Output Voltage Temperature Characteristics | $\frac{\Delta V_{OUT}}{\Delta T_{opr} \cdot V_{OUT}}$ | I <sub>OUT</sub> =30mA<br>-40 °C ≤ T <sub>opr</sub> ≤ 85 °C                                                                                                         | -      | ±100                                | -      | ppm/<br>°C | ①       |
| Short Circuit Current                      | I <sub>short</sub>                                    | V <sub>IN</sub> =V <sub>OUT</sub> +1.5V, V <sub>OUT</sub> =V <sub>SS</sub>                                                                                          | -      | E-6                                 | -      | mA         | ①       |

#### NOTE:

\* 1 Unless otherwise stated, V<sub>IN</sub> = V<sub>OUT(T)</sub> + 1.0V

\* 2 V<sub>OUT(T)</sub> :Nominal voltage

\* 3 V<sub>OUT(E)</sub> :Effective output voltage (I.e. The output voltage when "V<sub>OUT(T)</sub>+1.0V" is provided at the V<sub>IN</sub> pin while maintaining a certain I<sub>OUT</sub> value.)

\* 4 For output voltage accuracy, Please refer to E-1 table.

\* 5 V<sub>dif</sub> =V<sub>IN1</sub> -V<sub>OUT1</sub>

V<sub>OUT1</sub> :A voltage equal to 98% of the output voltage whenever an amply stabilized {V<sub>OUT(T)</sub> + 1.0V} is input with each I<sub>OUT</sub>.

V<sub>IN1</sub> :The input voltage when V<sub>OUT1</sub> appears as input voltage is gradually decreased.

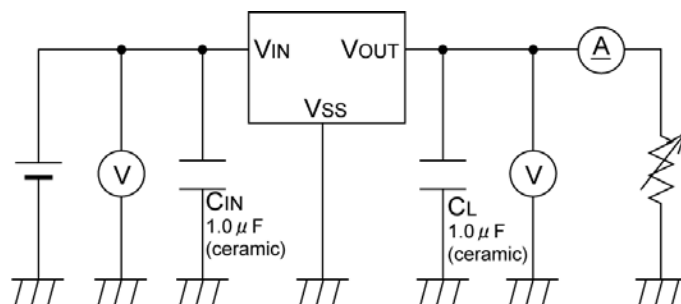
## ■ ELECTRICAL CHARACTERISTICS (Continued)

### ● Electrical Characteristics Chart

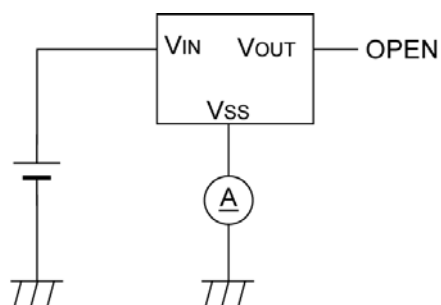
|                                     | E-1            |       |                |       | E-2                       | E-3                    | E-4                  |      | E-5                  |      | E-6              |
|-------------------------------------|----------------|-------|----------------|-------|---------------------------|------------------------|----------------------|------|----------------------|------|------------------|
| PARAMETER<br><br>NOMINAL<br>VOLTAGE | OUTPUT VOLTAGE |       |                |       | MAX.<br>OUTPUT<br>CURRENT | LOAD<br>REGULATIO<br>N | DROPOUT<br>VOLTAGE 1 |      | DROPOUT<br>VOLTAGE 2 |      | SHORT<br>CURRENT |
|                                     | 2%<br>ACCURACY |       | 1%<br>ACCURACY |       |                           |                        |                      |      |                      |      |                  |
| VOUT(T)                             | VOUT(E) (V)    |       | VOUT(E) (V)    |       | IOUTMAX (mA)              | ΔVOUT<br>(mV)          | Vdif1<br>(mV)        |      | Vdif2<br>(mV)        |      | Ishort<br>(mA)   |
|                                     | MIN.           | MAX.  | MIN.           | MAX.  | MIN.                      | MAX.                   | TYP.                 | MAX. | TYP.                 | MAX. | TYP.             |
| 1.2                                 | 1.170          | 1.230 | Not Available  |       | 60                        | 40                     | 460                  | 760  | 700                  | 960  | 180              |
| 1.3                                 | 1.270          | 1.330 |                |       |                           |                        | 400                  | 650  |                      |      |                  |
| 1.4                                 | 1.370          | 1.430 |                |       |                           |                        | 350                  | 590  | 580                  | 860  |                  |
| 1.5                                 | 1.470          | 1.530 |                |       | 300                       | 510                    | 450                  | 810  |                      |      | 155              |
| 1.6                                 | 1.568          | 1.632 |                |       | 250                       | 450                    |                      |      |                      |      |                  |
| 1.7                                 | 1.666          | 1.734 |                |       | 200                       | 410                    |                      |      |                      |      |                  |
| 1.8                                 | 1.764          | 1.836 |                |       | 150                       | 390                    | 100                  | 370  | 350                  | 710  | 130              |
| 1.9                                 | 1.862          | 1.938 |                |       |                           |                        |                      |      |                      |      |                  |
| 2.0                                 | 1.960          | 2.040 | 1.980          | 2.020 |                           |                        |                      |      |                      |      |                  |
| 2.1                                 | 2.058          | 2.042 | 2.079          | 2.121 |                           |                        |                      |      |                      |      |                  |
| 2.2                                 | 2.156          | 2.244 | 2.178          | 2.222 |                           |                        |                      |      |                      |      |                  |
| 2.3                                 | 2.254          | 2.346 | 2.277          | 2.323 |                           |                        |                      |      |                      |      |                  |
| 2.4                                 | 2.352          | 2.448 | 2.376          | 2.424 | 150                       | 55                     |                      |      |                      |      | 75               |
| 2.5                                 | 2.450          | 2.550 | 2.475          | 2.525 |                           |                        |                      |      |                      |      |                  |
| 2.6                                 | 2.548          | 2.652 | 2.574          | 2.626 |                           |                        |                      |      |                      |      |                  |
| 2.7                                 | 2.646          | 2.754 | 2.673          | 2.727 |                           |                        |                      |      |                      |      |                  |
| 2.8                                 | 2.744          | 2.856 | 2.772          | 2.828 |                           |                        |                      |      |                      |      |                  |
| 2.9                                 | 2.842          | 2.958 | 2.871          | 2.929 |                           |                        | 200                  | 60   | 75                   | 350  |                  |
| 3.0                                 | 2.940          | 3.060 | 2.970          | 3.030 |                           |                        |                      |      |                      |      |                  |
| 3.1                                 | 3.038          | 3.162 | 3.069          | 3.131 |                           |                        |                      |      |                      |      |                  |
| 3.2                                 | 3.136          | 3.264 | 3.168          | 3.232 |                           |                        |                      |      |                      |      |                  |
| 3.3                                 | 3.234          | 3.366 | 3.267          | 3.333 |                           |                        |                      |      |                      |      |                  |
| 3.4                                 | 3.332          | 3.468 | 3.366          | 3.434 |                           |                        |                      |      |                      |      |                  |
| 3.5                                 | 3.430          | 3.570 | 3.465          | 3.535 |                           |                        |                      |      |                      |      |                  |
| 3.6                                 | 3.528          | 3.672 | 3.564          | 3.636 |                           |                        |                      |      |                      |      |                  |
| 3.7                                 | 3.626          | 3.774 | 3.663          | 3.737 |                           |                        |                      |      |                      |      |                  |
| 3.8                                 | 3.724          | 3.876 | 3.762          | 3.838 |                           |                        |                      |      |                      |      |                  |
| 3.9                                 | 3.822          | 3.978 | 3.861          | 3.939 | 250                       | 70                     | 60                   | 320  | 200                  | 630  |                  |
| 4.0                                 | 3.920          | 4.080 | 3.960          | 4.040 |                           |                        |                      |      |                      |      |                  |
| 4.1                                 | 4.018          | 4.182 | 4.059          | 4.141 |                           |                        |                      |      |                      |      |                  |
| 4.2                                 | 4.116          | 4.284 | 4.158          | 4.242 |                           |                        |                      |      |                      |      |                  |
| 4.3                                 | 4.214          | 4.386 | 4.257          | 4.343 |                           |                        |                      |      |                      |      |                  |
| 4.4                                 | 4.312          | 4.488 | 4.356          | 4.444 |                           |                        |                      |      |                      |      |                  |
| 4.5                                 | 4.410          | 4.590 | 4.455          | 4.545 |                           |                        |                      |      |                      |      |                  |
| 4.6                                 | 4.508          | 4.692 | 4.554          | 4.646 |                           |                        |                      |      |                      |      |                  |
| 4.7                                 | 4.606          | 4.794 | 4.653          | 4.747 |                           |                        |                      |      |                      |      |                  |
| 4.8                                 | 4.704          | 4.896 | 4.752          | 4.848 |                           |                        |                      |      |                      |      |                  |
| 4.9                                 | 4.802          | 4.998 | 4.851          | 4.949 | 80                        | 50                     | 290                  | 175  | 600                  |      |                  |
| 5.0                                 | 4.900          | 5.100 | 4.950          | 5.050 |                           |                        |                      |      |                      |      |                  |

## TEST CIRCUITS

Circuit ①

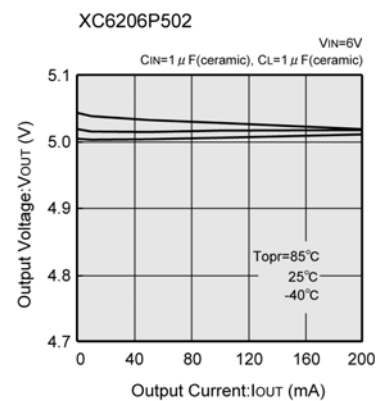
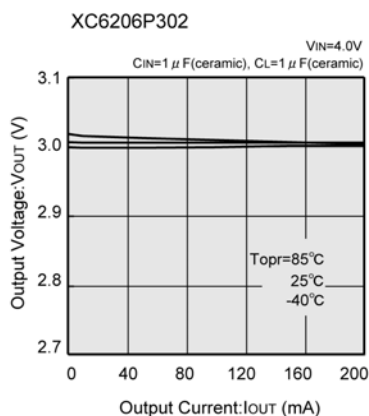
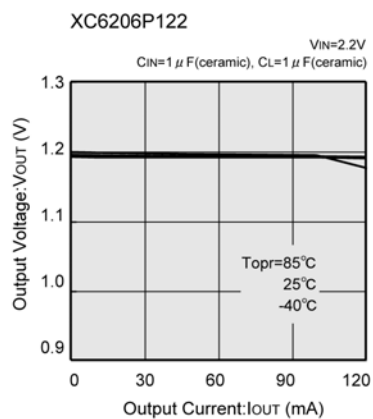


Circuit ②

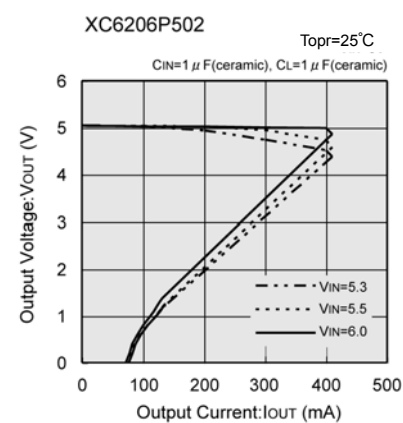
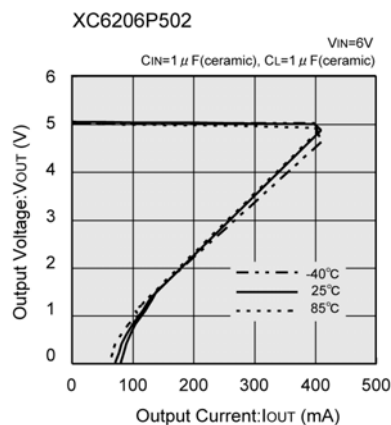
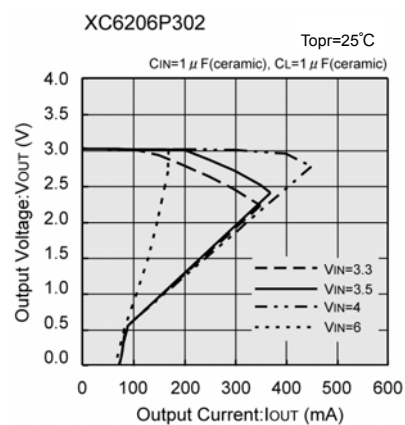
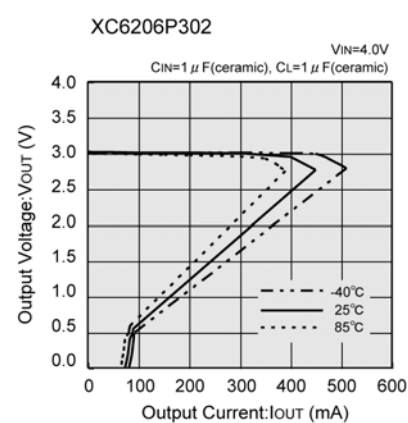
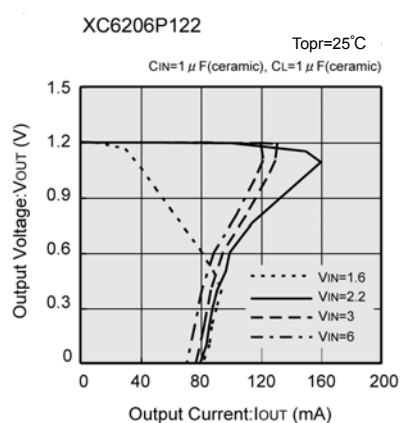
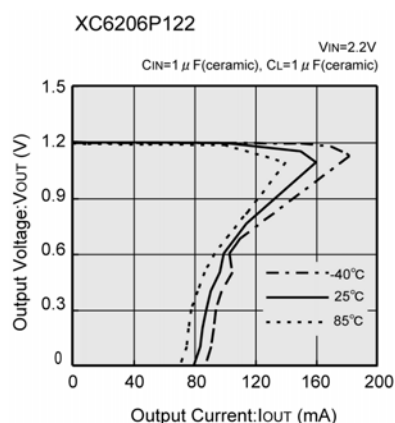


## ■ TYPICAL PERFORMANCE CHARACTERISTICS

### (1) Output Voltage vs. Output Current

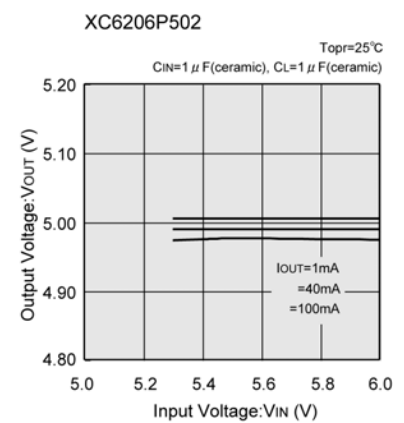
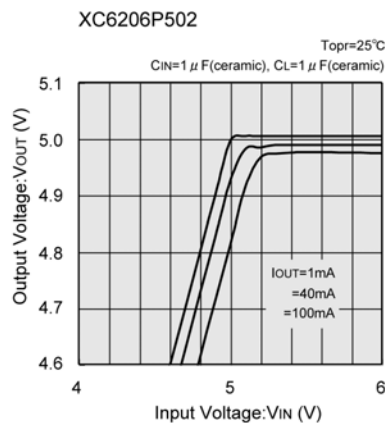
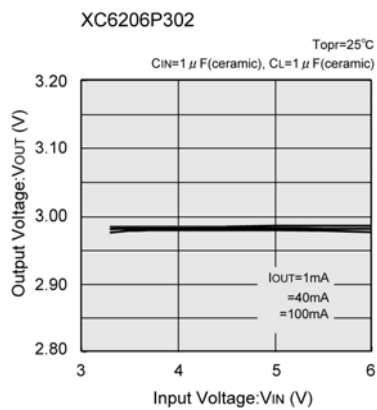
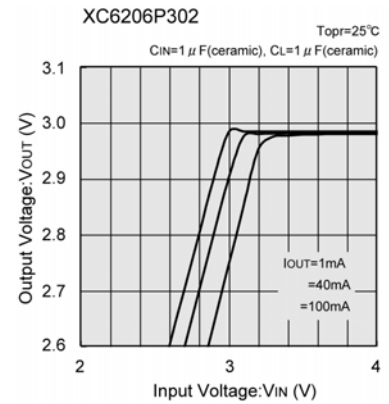
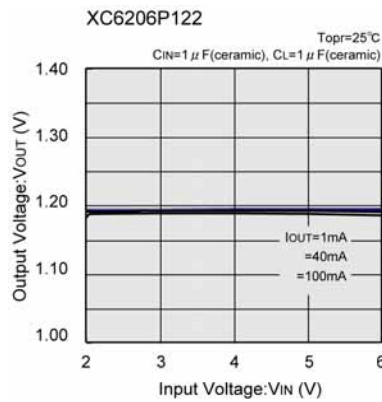
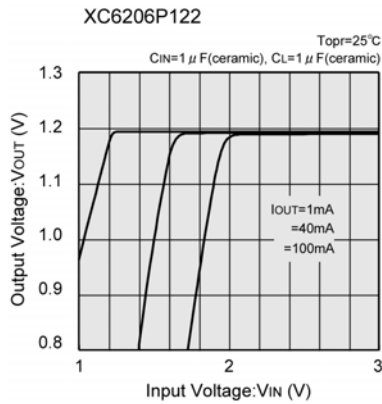


### (2) Current Limit

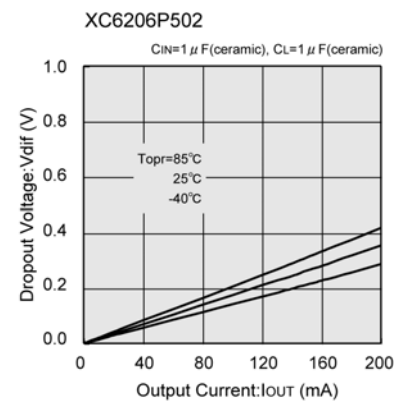
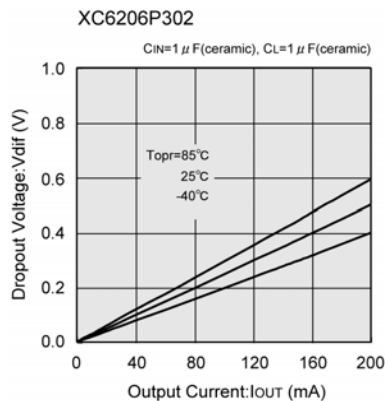
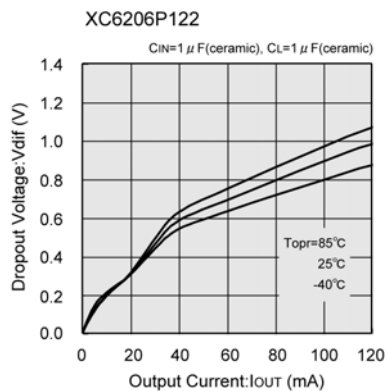


## TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (3) Output Voltage vs. Input Voltage

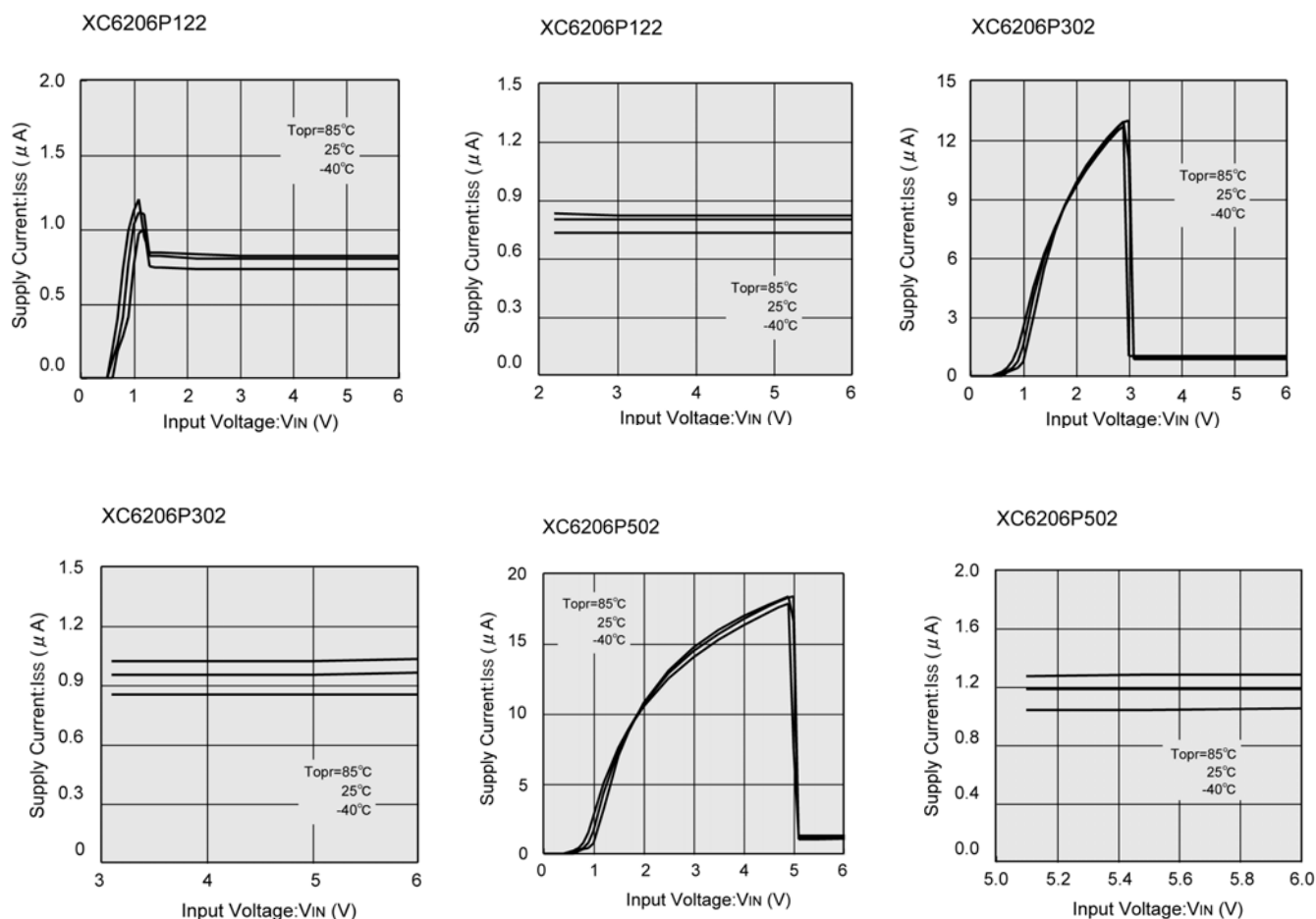


### (4) Dropout Voltage vs. Output Current

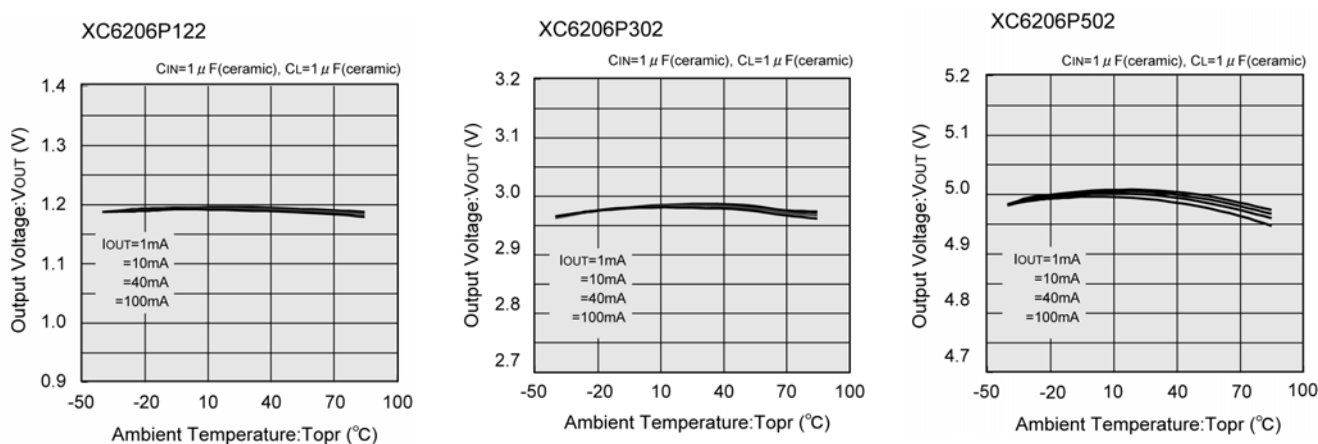


## ■ TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (5) Supply Current vs. Input Voltage

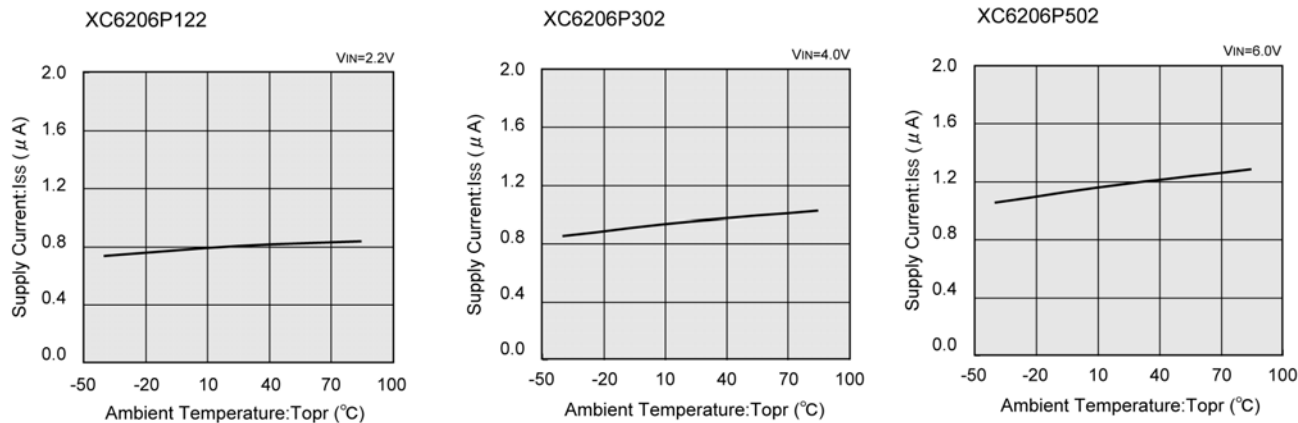


### (6) Output Voltage vs. Ambient Temperature

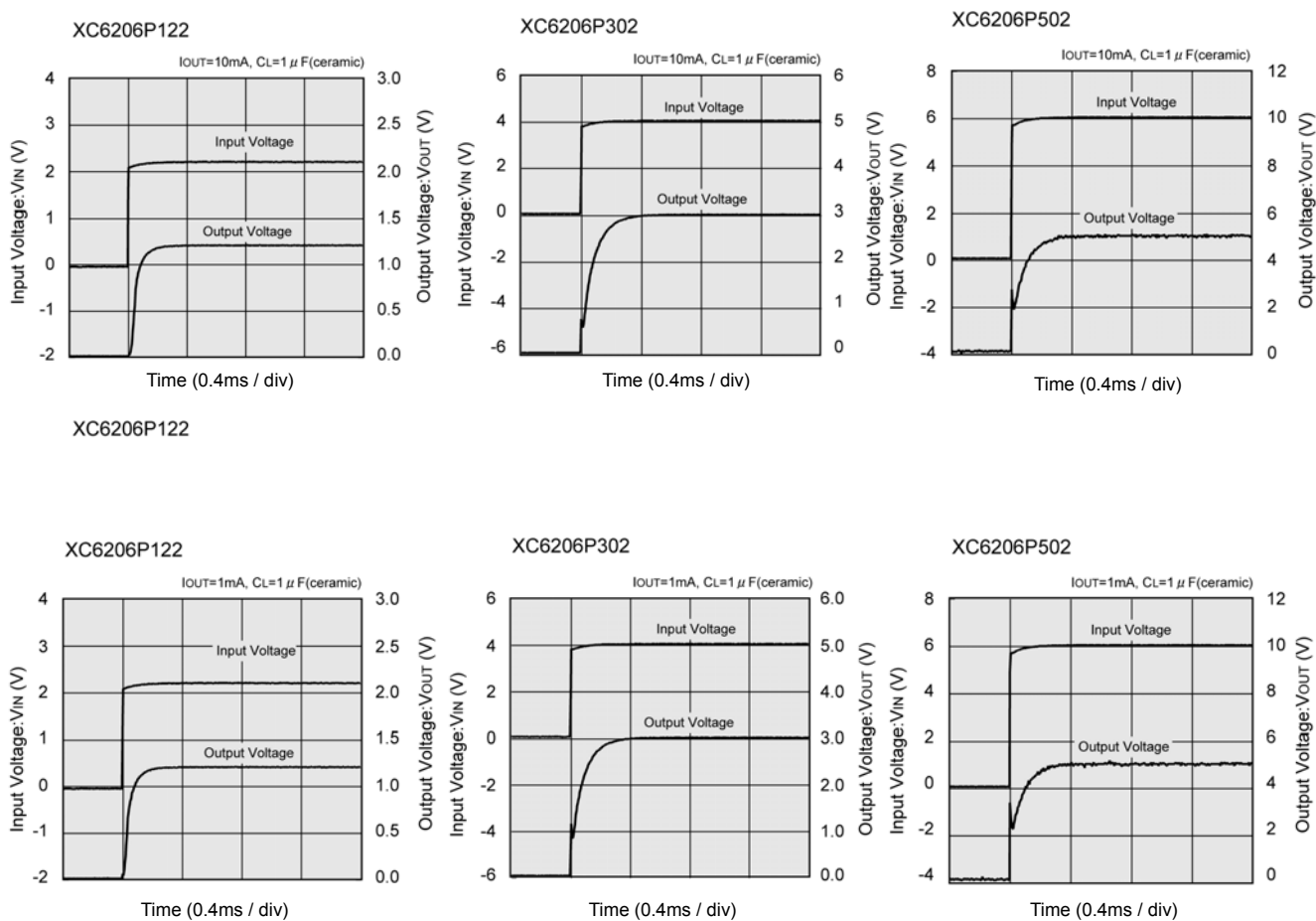


## TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (7) Output Voltage vs. Ambient Temperature

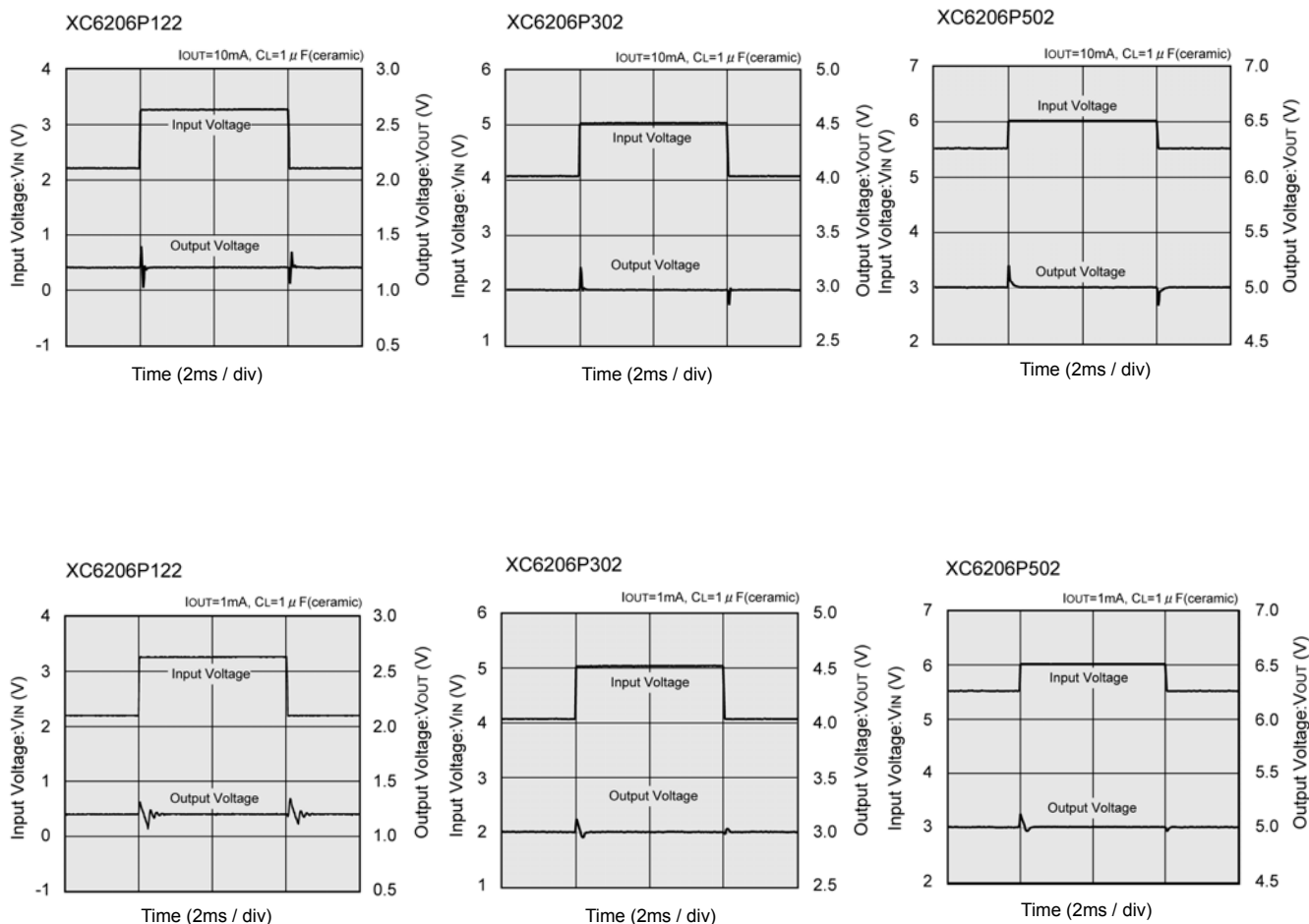


### (8) Input Transient Response 1

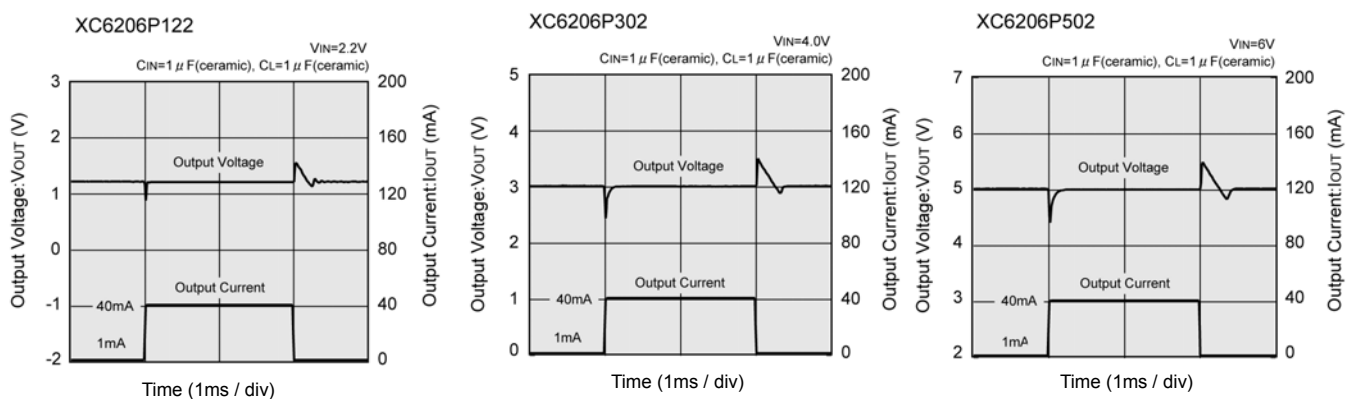


## ■ TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

### (9) Input Transient Response 2

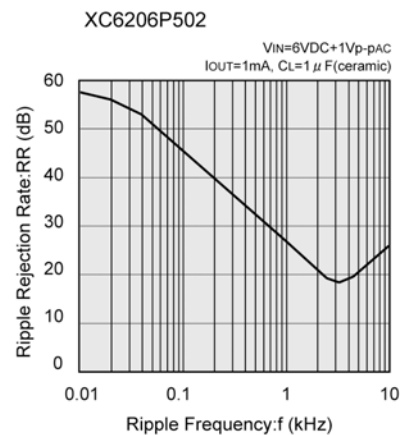
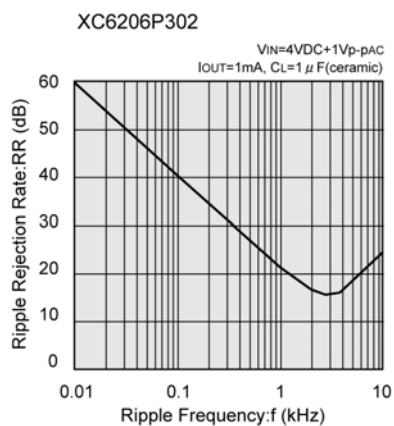
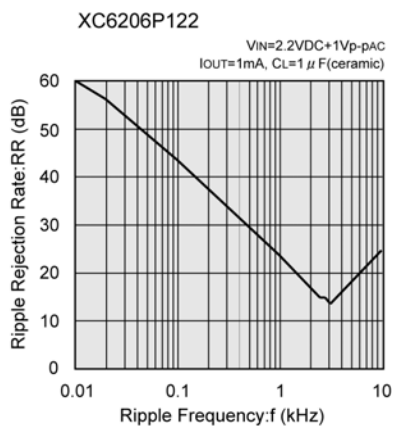
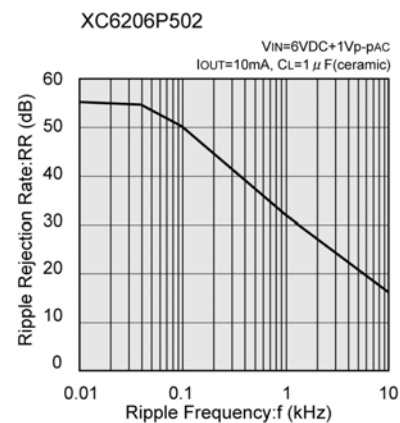
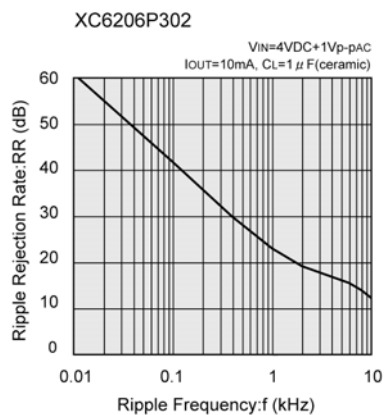
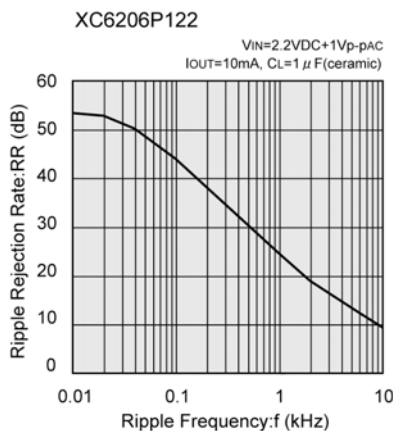


### (10) Load Transient Response



## TYPICAL PERFORMANCE CHARACTERISTICS (Continued)

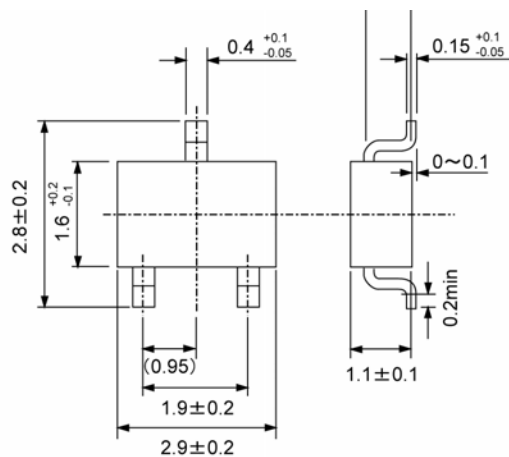
### (11) Ripple Rejection Rate



## ■ PACKAGING INFORMATION

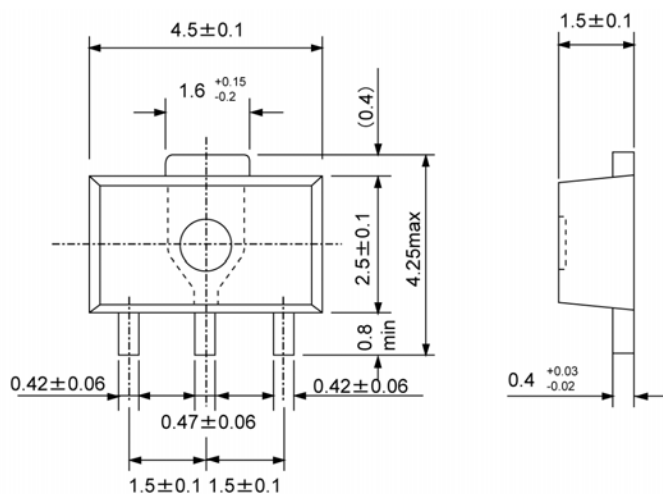
### ● SOT-23

Unit : mm



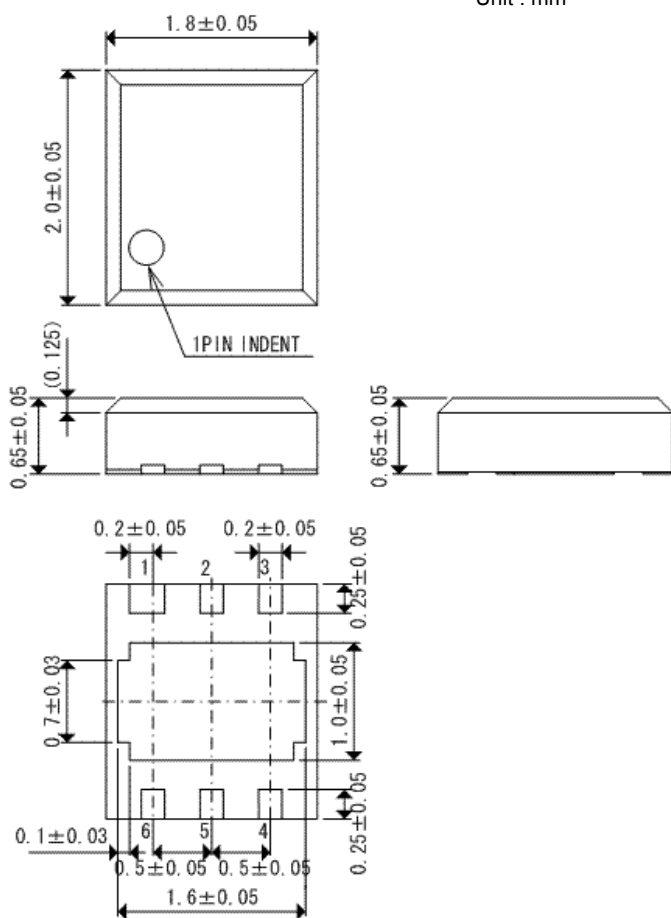
### ● SOT-89

Unit : mm



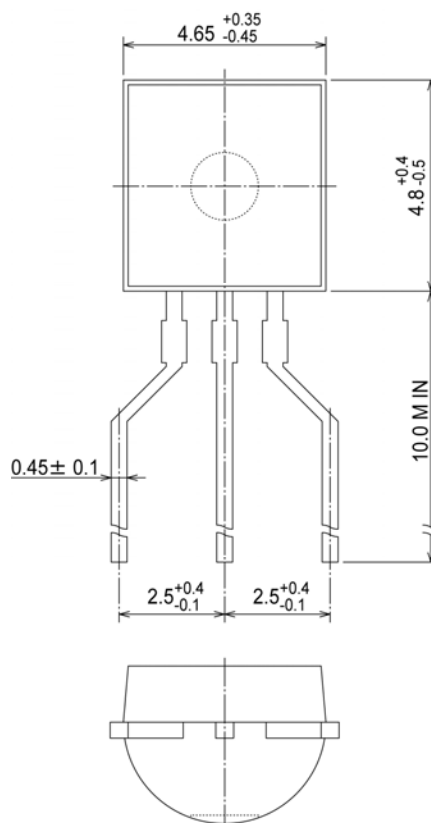
### ● USP-6B

Unit : mm



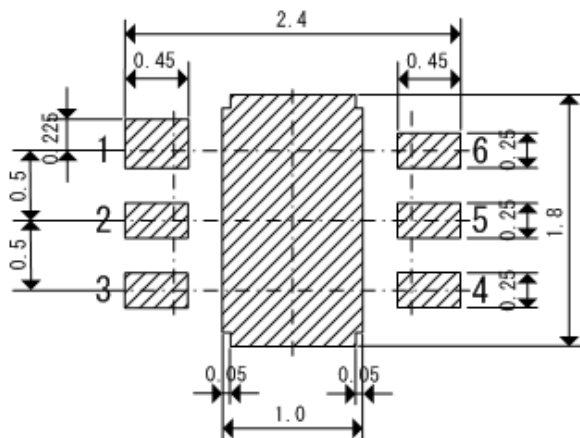
### ● TO-92

Unit : mm

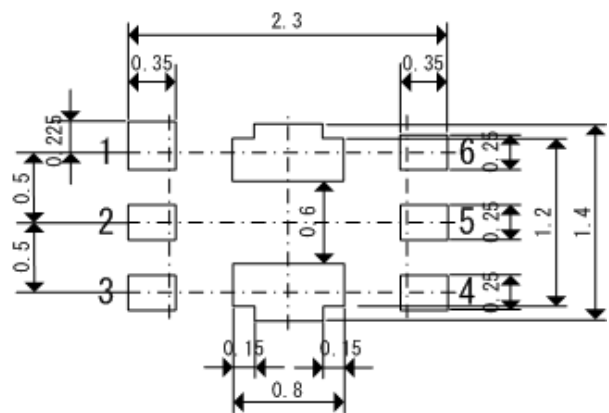


## PACKAGING INFORMATION (Continued)

● USP-6B Reference Pattern Layout

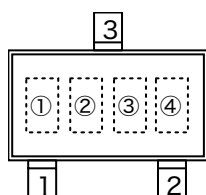


● USP-6B Reference Metal Mask Design

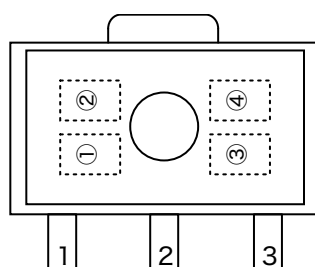


## MARKING RULE

### ●SOT-23, SOT-89



SOT-23  
(TOP VIEW)



SOT-89  
(TOP VIEW)

① represents product number

| MARK | PRODUCT SERIES |
|------|----------------|
| 6    | XC6206P*****   |

② represents 3 pins regulator

| MARK               |                    | PRODUCT SERIES |
|--------------------|--------------------|----------------|
| VOLTAGE=0.1 ~ 3.0V | VOLTAGE=3.1 ~ 6.0V |                |
| 5                  | 6                  | XC6206P*****   |

③ represents output voltage

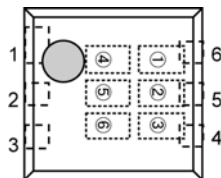
| MARK | VOLTAGE (V) |     |   | MARK | OUTPUT VOLTAGE (V) |     |   |
|------|-------------|-----|---|------|--------------------|-----|---|
| 0    | -           | 3.1 | - | F    | 1.6                | 4.6 | - |
| 1    | -           | 3.2 | - | H    | 1.7                | 4.7 | - |
| 2    | -           | 3.3 | - | K    | 1.8                | 4.8 | - |
| 3    | -           | 3.4 | - | L    | 1.9                | 4.9 | - |
| 4    | -           | 3.5 | - | M    | 2.0                | 5.0 | - |
| 5    | -           | 3.6 | - | N    | 2.1                | -   | - |
| 6    | -           | 3.7 | - | P    | 2.2                | -   | - |
| 7    | -           | 3.8 | - | R    | 2.3                | -   | - |
| 8    | -           | 3.9 | - | S    | 2.4                | -   | - |
| 9    | -           | 4.0 | - | T    | 2.5                | -   | - |
| A    |             | 4.1 | - | U    | 2.6                | -   | - |
| B    | 1.2         | 4.2 | - | V    | 2.7                | -   | - |
| C    | 1.3         | 4.3 | - | X    | 2.8                | -   | - |
| D    | 1.4         | 4.4 | - | Y    | 2.9                | -   | - |
| E    | 1.5         | 4.5 | - | Z    | 3.0                | -   | - |

④ represents production lot number

0 to 9, A to Z, and inverted 0 to 9, A to Z repeated. (G, I, J, O, Q, W excepted.)

## MARKING RULE (Continued)

### ● USP-6B



USP-6B  
(TOP VIEW)

①② represents product number

| MARK |   | PRODUCT SERIES |
|------|---|----------------|
| ①    | ② |                |
| 0    | 6 | XC6206P***D*   |

③ represents 3 pins regulator

| MARK | PRODUCT SERIES |
|------|----------------|
| P    | XC6206P***D*   |

④⑤ represents output voltage

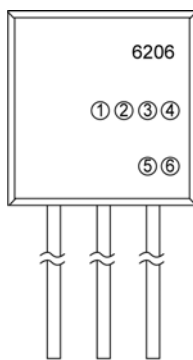
| MARK |   | OUTPUT VOLTAGE(V) | PRODUCT SERIES |
|------|---|-------------------|----------------|
| ④    | ⑤ |                   |                |
| 3    | 3 | 3.3               | XC6206P33*D*   |
| 5    | 0 | 5.0               | XC6206P50*D*   |

⑥ represents production lot number

0 to 9, A to Z repeated. (G, I, J, O, Q, W excluded)

\*No character inversion used.

### ● TO-92



TO-92 (T Type)  
(SIDE VIEW)

① represents type of regulator

| MARK | PRODUCT SERIES |
|------|----------------|
| P    | XC6206P*****   |

②③ represents output voltage

| MARK |   | VOLTAGE (V) | PRODUCT SERIES |
|------|---|-------------|----------------|
| ②    | ③ |             |                |
| 3    | 3 | 3.3         | XC6206P33***   |
| 5    | 0 | 5           | XC6206P50***   |

④ represents output voltage accuracy

| MARK | OUTPUT VOLTAGE ACCURACY | PRODUCT SERIES |
|------|-------------------------|----------------|
| 1    | ± 1%                    | XC6206P**1**   |
| 2    | ± 2%                    | XC6206P**2**   |

⑤ represents least significant digit of the production year

| MARK | PRODUCTION YEAR |
|------|-----------------|
| 3    | 2003            |
| 4    | 2004            |

⑥ represents production lot number

0 to 9, A to Z repeated. (G, I, J, O, Q, W excluded)

\*No character inversion used.

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